CHRISTOPHER GROUNDS

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1 Summary

I am a highly experienced software engineer & software engineering manager who has worked on a range of different domains, problems, and technologies. My main focus throughout my career has been on excellence and best practices - both cultural and engineering.

This is evidenced by my previous and current roles, where I have promoted and encouraged well-tested, clean code through a combination of fast-feedback loops, TDD, best practices such as pair programming, and creating a safe working environment. Through this focus, I have been instrumental in shaping and forming positive team culture and engineering best practices, whilst taking lead on green-field projects critical to achieving business and organisational goals

In my current role, I have been instrumental in shaping and forming positive team culture and engineering best practices, whilst taking lead on green-field projects critical to achieving business and organizational goals.

My ambition is to help shape agile teams, to enable them to innovate and move quickly. I believe this is best achieved through building autonomous teams, deeply routed in extreme programming best practices, with emphasis on automation through CI/CD, pair programming involving all relevant disciplines, coupled with expert knowledge of the technologies involved.

I bring extremely strong technical abilities, ranging from cloud, web, mobile, and the full software-engineering stack - from strongly-typed functional programming languages (such as Haskell and TypeScript) to web-standards and design. I have had considerable influence on large-scale production applications and pride myself on the end-result achieved.

All in all, I view software as a team enterprise and I am a believer in the value of the principles of extreme programming. This involves a strong emphasis on pair/mob programming and including all disciplines in the software-creation process as early as possible, coupled with expert knowledge of the technologies involved.

Core competencies: Rust, React, Node, TypeScript, AWS Serverless stack, Agile best practices and engineering excellence (TDD, CI/CD, XP practices), functional programming (Haskell, Rust)

2 Experience

01/2025 - present **Software Engineering Manager, Switch2 Energy**

In my role at Switch2 I have been responsible for setting the technical direction and strategy for the company as the heat network industry transitions into a significant part of the UK heat grid. My role has been multi-faceted in focus, covering both software and data, and I have worked with a range of stakeholders, ranging from line management of software engineers to senior leadership to set future direction and strategy.

I my role I have:

- Shifted the engineering department mindset from project-led to product-orientated thinking, focusing the team and external stakeholders on streams of products rather than short-term projects, to achieve more successful user outcomes
- Created and implemented a new data strategy, migrating bespoke data pipelines to Databricks. This involved working with SLT, data stakeholders, and data/software engineers to align expectations/requirements. The migration to Databricks lead to creation of new data streams and reports that were impossible beforehand, unlocking new value for the business
- Migrated part of the stack to Rust. The outcome was 3-fold: greater software quality with fewer bugs, higher software performance/lower cost, and increased developer productivity and morale

- As hiring manager brought in 2 new engineers to accelerate growth. My role was responsible for building the business case for the new engineers, interviewing, and orchestrating with SLT to ensure successful acquisition
- Successfully delivered multiple products and projects: integration our new G10 devices into our business estate, delivering value from our data platform, integrating external third-party devices into our business estate, building a new user-facing Optimise platform, building new internal tooling to automate and ease internal processes
- Supported the development of software engineering expertise in the team through 1:1 mentoring and encouragement of a team-first sharing approach
- Re-architected and designed the frontend stack to be component-driven

Tools & Techs: Pair programming, Test & TypeDD, Rust, Node, React, AWS, CI/CD (Github Actions)

04/2023 - 01/2025 Development Team Lead, Switch2 Energy

As the team lead for the Switch2 software development team, I have focused the team on delivering value and improving the value flywheel, shaped team culture and practice, and led the team to achieve key business goals. I have taken a team-first approach to support the team in the development and delivery of business requirements, utilising a variety of techniques and best practices from agile and XP.

I have managed a team of 3 senior software engineers and 2 PowerBI engineers. My responsibilities have involved line-management and mentoring, strategy, architecture, and hands-on coding. I have made long-term, strategic decisions around the technical roadmap and worked with senior stakeholders to ensure understanding and support of the technical roadmap.

I my role I have:

- Led and facilitated event-storming sessions, to create shared understanding and ubiquitous language in the team. This has helped de-silo product knowledge and create team focus
- Led and facilitated Wardley mapping sessions, to help stakeholders achieve strategic alignment and better visualise and understand their context and landscape
- Delivered critical business systems
- Supported the development of software engineering expertise in the team through 1:1 mentoring and encouragement of a team-first sharing approach
- Led and architected a data migration from an on-premise data warehousing solution to Databricks, thereby unlocking large amounts of data processing and insight
- Engaged with senior stakeholders to re-organise and restructure the team to align it with industry software standards
- Ensured successful collaboration with Switch2 R&D to achieve successful product launches
- Made appropriate production-use of LLMs to automate difficult data processing
- Overseen third-party contractors to ensure the timely delivery of work

Tools & Techs: Pair programming, Test & TypeDD, Rust, Node, React, AWS, CI/CD (Github Actions)

02/2021 - 04/2023 Staff Engineer, AND Digital

As a Staff Engineer I have been working closely with stakeholders and other engineers, bridging the gap between client and AND, ensuring successful delivery and execution of client requirements. I am completely focused on delivery user value and helping organisations accelerate their digital ambitions.

Over my time at AND I have helped lead three projects to successfully meet deliverables and deliver value. I've worked closely with client stakeholders, AND delivery leads, and other developers to ensure work is completed as smoothly and efficiently as possible. Off client, I have contributed to the culture of the developer community, mentored other developers (formally and through more ad-hoc relationships), and acted as a positive senior role model.

I have worn many hats at AND, showing flexibility and pragmatism, with a focus on enabling each team to deliver to the best of their abilities, all whilst being pivotal in helping develop the culture and best practices across the whole of the Jemison AND club. I have focused on engineering and agile best practices and see my self as a cross-cutting generalist who helps organisations and teams deliver value at pace.

My roles:

- The Very Group - Staff Software Engineer
- Princes' Trust - Senior Software Engineer
- Internal Project - Lead Software Engineer
- Boohoo - Led workshops

On my current client engagement at The Very Group I have helped lead 2 teams to the successful delivery of 2 phases of work. I have helped design and implement an **AWS event-driven, serverless** architecture, that serves as a piece of critical infrastructure that consumes and produces financial events. This work has allowed The Very Group to migrate from a previous service provider, allowing the business to offer better services to customers. I also was instrumental in designing and implementing the **automation test strategy** (**E2E and integration tests**), have **mentored** junior developers, and practiced and preached **test-driven design** and **continuous integration**. I have also been seconded to help other teams improve **performance and resilience** of their AWS infrastructure and have contributed to the Very **React** frontend consuming our serverless services.

Some feedback I've recevied:

> A brilliant team player, coaching associates, going the extra mile, escalating appropriately with empathy, diplomatic. & an all round great guy
>
> A reliable team player who I can trust to solve problems with minimal fuss. Excellent communicator, & know Chris will get the best possible outcome
>
> mob programming sessions was great. It was easy to discuss different implementation techniques [with Chris]

Tools & Techs: Pair programming, TDD, Node, React, AWS, CI/CD, Jenkins, JavaScript, E2E Testing

02/2021 - present Senior Software Engineer, AND Digital

Using my industry experience to help organisations deliver, execute, and accelerate their digital ambitions.

I've been active in recruitment, bringing talent and the right skill-sets into AND, and decisive in creating a community of practice within AND. On and off-client I have mentored and helped develop other ANDis, championing engineering best practices, instilling technical and non-technical knowledge, and agile best practices such as XP.

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Tools & Techs: Pair programming, TDD, Node, React, AWS, CI/CD, Jenkins, JavaScript, E2E Testing, Contentful

02/2021 - present Senior Software Engineer, BBC

Significant contribution to the technical design and decision making on a wide range of components across the Universal App Platform, a Capacitor hybrid app platform, including build system and toolchains. Instrumental in forming team values and ways of working and taken lead on a wide range of projects.

Continuation of my work on the Universal App Platform; responsible for delivery of the beta and live app releases.

Tools & Techs: Pair programming, TDD, Capacitor, React, Trunk-based dev, AWS, CI/CD, Jenkins, JavaScript, E2E Testing

03/2020 - 02/2021 **Software Engineer, BBC**

Significant contribution to the technical design and decision making on a wide range of components across the Universal App Platform, a Capacitor hybrid app platform, including build system and toolchains. Instrumental in forming team values and ways of working and taken lead on a wide range of projects.

Tools & Techs: Pair programming, TDD, Capacitor, React, Trunk-based dev, AWS, CI/CD, Jenkins, JavaScript, E2E Testing

11/2018 - 03/2020 **Software Engineer, Tracsis**

Software Engineer at Tracsis working on the resource management system of TRACS Enterprise, a high-availability suite of software, responsible for building high-quality and thoroughly-tested software to help train operating companies' processes become more efficient and streamlined.

Throughout my time here I have continuously encouraged and promoted agile best practices such as TDD, pair-programming, kanban-practices, CD, and feature-toggled based development.

Tools & Techs: Pair programming, TDD, Trunk-based dev, Haskell, AWS, CI/CD, Jenkins, JavaScript, E2E Testing

12/2017 - 11/2018 **Junior Software Engineer, BBC**

A member of the Newsround team, working with React and the BBC's proprietary serverless platform to build a high-performing new website for Newsround, transitioning it to a digital future.

Further to the Newsround website, I have built internal tooling to facilitate continuous deployment, championed functional programming and safe code, given talks on Haskell, lead and organised a Haskell study group, and been involved with assessment centres, bringing talent into the BBC.

Tools & Techs: Node.js/React, Elm, TypeScript, TDD, Pair Programming, CI/CD, Serverless

06/2016 - 12/2017 **Web Software Developer, Age Partnership**

A member of the web team, responsible for writing greenfield software and web applications, maintaining legacy code, and responding to operational problems. Also the unit testing project lead responsible for unit testing the codebase.

3 Education

09/2015 - 09/2016 **MA. (Postgraduate) Linguistics with a Specialisation in Syntax**
University College London (UCL)
Degree Classification: *Distinction*

09/2012 - 06/2015 **MA. (Undergraduate) (Hons) Language & Linguistics and Philsophy**
University of Aberdeen
Degree Classification: *First Class, with Honours*

4 Open Source

- GHC: Fixed GHC.Natural documentation typo
- Learn You a Haskell companion exercises
- Vim-one: added Haskell support
- Intero-neovim: Added functionality to customize Intero window size and position

5 Personal Projects (selection; all available on github)

- Haskell random name generator, deployed on Heroku and AWS Lambda

- Halendar: terminal based calendar (Haskell)
- Htex: HTML-LaTeX compiler (Haskell)
- Multiprocessing TCP portscanner (Python)
- Interactive calculator REPL (Haskell)

6 Presentations & Talks

- 18/02/2019, 'Haskell at Tracsis: some lessons', *Manchester Lambda Lounge*: Manchester
- 22/11/2018, 'Pair programming', *Tracsis*: Leeds
- 16/08/2018, 'Using Monoids to build UI', *Manchester Lambda Lounge*: Manchester
- 23/03/2018, 'Haskell: programming purity', *BBC*: MediaCity